Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI & Autonomous Agents · Python / LangGraph / AWS · Large Language Model (LLM) Systems & Evaluation

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SUMMARY

AI Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source evaluation and monitoring platform for production AI agents: CI-integrated regression gate, LLM-as-judge scoring with full agent tracing, multi-trial pass@k harness, and hallucination detection -- built to operate agentic AI responsibly at scale. Built Leasel -- an 8-agent LangGraph system with retrieval-augmented generation (RAG), tool calling, HITL guardrails, and structured data extraction across 1,240 test cases (93.6% precision). Regulated-domain production experience: AI on FHIR (HIPAA-compliant PHI system with RBAC) + Entrata PHI compliance (zero security incidents). Seven years shipping enterprise Python backend systems under compliance and reliability constraints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

Agentic AI & LLM Engineering: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Semantic Search, Tool Calling, Function Calling, Prompt Engineering, Schema Validation, Structured Outputs, HITL, Guardrails, Hallucination Detection, Output Validation

AI Evaluation & Monitoring: LLM Evaluation, Evaluation Pipelines, LLM-as-Judge, CI/CD Eval Gate, Pass@k Harness, Agent Tracing, Production Monitoring, Regression Suites, Quality Metrics, Root Cause Analysis, Token Optimization, Reliability Engineering, MLOps, LLMOps

Languages & Frameworks: Python, TypeScript, JavaScript, Java, SQL, FastAPI, REST APIs, React.js, Flask, Angular, Node.js, Bash, HTML/CSS

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), Event-Driven Architecture (RabbitMQ async), Redis, Microservices, HIPAA Compliance, PHI Data Handling, PostgreSQL, Oracle, Git, Linux, Agile/Scrum, OpenAI API, Anthropic API, Groq, Ollama

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic system with RAG retrieval, guardrails, and failure handling for an external client -- owned design through delivery including LLM selection, orchestration architecture, and validation layer; delivered with all compliance decisions cited to exact source.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · enterprise production systems)
Sep 2018 - Aug 2024

Senior Software Engineer

- **Owned compliance-critical data systems with zero tolerance for errors** -- annual HUD income-limit pipeline (180,000 domain records, housing eligibility for 100K+ residents); zero compliance violations across 7 years; baseline validation, reconciliation, and rollback preparation on every release. Regulated domain discipline: reliability and auditability built into every system.
- Led AWS S3 document migration -- 100% PHI compliance with Libsodium-encrypted pipelines, zero security incidents; designed and owned HIPAA-adjacent data workflows throughout tenure.
- Shipped 20+ full-stack production modules (Python, PostgreSQL, TypeScript/JavaScript, RabbitMQ event-driven async) across 120+ Agile sprints; built Python/Bash automation pipelines eliminating 80% of manual workload; optimized 200+ SQL queries (30% performance gain).
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools reducing investigation time by 40%.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data integrity checks across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Evaluation & Monitoring Platform for Production Agents
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source evaluation and monitoring platform for production AI agents -- CI-integrated regression gate (quality failures blocked before deployment), LLM-as-judge scoring with full agent tracing (every failure has a diagnosable root cause), multi-trial pass@k harness measuring whether changes improve agent behavior over time. Built to operate agentic AI responsibly at scale.
- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama); 15-endpoint FastAPI REST backend; evaluated Leasel across 1,240 production-quality test cases.

Leasel: 8-Agent LangGraph Agentic System with RAG & HITL Guardrails
2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic system: 8 LangGraph agents with conditional routing, tool calling, retrieval-augmented generation over compliance documents, structured data extraction, HITL escalation on regulated decisions -- with reliability and auditability built into every workflow step.

- 93.6% precision, 87.9% F1, pass@1 81.7% across 1,240 test cases -- with full observability through AgentProbe eval harness.

AI on FHIR: HIPAA-Compliant Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, NLP-to-FHIR API translation, LLM + BERT integration; security and data privacy built in from architecture; reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017